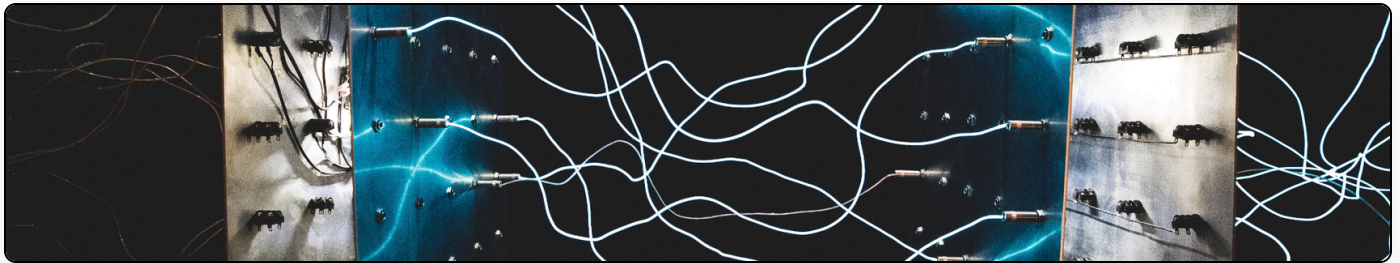


What if downloads are not actually non-rivalrous?

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One of the basic tenets of digital philosophy is that downloads are non-rivalrous. When I download something it remains accessible to other users to download. But what if this isn't actually true and what if there is a strong disutility premise at work?

Bitcoin has not done much for the world, but it does provide a useful reference point for thinking about this. The annual energy consumption of Bitcoin is enormous. Estimated in 2020 at 181.07 terawatt hours, Bitcoin has power consumption equal to that used by Thailand and a carbon footprint the size of Bangladesh, despite processing just a few hundred million transactions, compared to traditional processors which were in the realm of several hundred billion payments. A single transaction on this blockchain is equal to 62 days of energy usage by the average U.S. household, while it takes the carbon footprint of almost 2 million traditional Visa credit card purchases to make just one spend. The network also creates a huge amount of e-waste as dedicated mining rigs burn out under extreme usage.

While this sounds like a flaw in cryptocurrency use, though, it is not; or, at least, it is a flaw by design. The important point to understand in cryptocurrency mining and operation is that without the external dependence on electricity, there is nothing staked to put value behind a particular digital coin. Indeed, as yanmaani puts it, the 'idea is to have a method to burn electricity in a provable way. The electricity used does not directly produce anything of value, except for a proof that (approximately) this amount of electricity was consumed by this or that node.' Electricity

works as a value proposition because it is ‘costly’ (you have to pay for it), ‘irreversible’ (once you’ve used the electricity, you have sunk the spend), and ‘self-certifying’ (the network can see that the proof-of-work cost a certain amount of computational energy, which correlates to electricity).

Hence, when people say that cryptocurrencies are mere betting or speculation, because they are not underwritten by any real-world scarce resource (and despite the fact that the post-Bretton Woods system of economics means that contemporary real currencies are also no longer linked to the gold standard), this couldn’t be further from the truth. There is a strong external dependence on computational power – and, therefore, electricity. Certainly, this has variable value implications as the cost of electricity and the way it is produced can be radically different between different regions of the world. Nonetheless, there is a scarce resource that underpins Bitcoin and this expenditure is ‘at stake.’

Every download on the internet has a computational cost. To transmit data around the world in the form of a download requires energy expenditure and infrastructure maintenance. The differential effects of climate change mean that to state that our use has no effect on others, when, at scale, this energy expenditure may lead to global warming and other catastrophes is simply therefore not correct, ethically.

The disutility principles for public goods are set out by William H. Oakland, who writes that “A public good may also create disutility or reduced profits. A cigarette smoked in a crowded classroom is an example. The disutility suffered by one non-smoker does not reduce the disutility suffered by other non-smokers. Similarly, the reduced profits suffered by one fisherman because of water pollution does not reduce the loss to other fishermen” (in Oakland, William H. ‘Theory of Public Goods’. In *Handbook of Public Economics*, edited by Alan J. Auerbach and Martin S. Feldstein, 2:485–535. *Handbooks in Economics*. Amsterdam: Elsevier, 1985. p. 485).

Clearly, there is a disutility principle at work in the idea of a download being a non-rivalrous artifact. In essence, there are costs to downloads that are not borne by the people who benefit from those downloads. In this sense, downloads are less public goods than distributionally-funded goods.